

OpenModelica Desktop App (PyQt6)

A simple desktop application built with Python and PyQt6 that launches an executable generated from an OpenModelica model. The app accepts an executable path, a start time, and a stop time, then runs the simulation from the GUI.

Project Overview

This project is based on the OpenModelica `TwoConnectedTanks` model. The model is compiled in OMEdit to generate a simulation executable and its related runtime files. A Python GUI is then used to launch that executable with user-provided simulation arguments.

Features

- Browse and select the OpenModelica executable
- Enter start time and stop time
- Validate simulation inputs
- Run the executable from a desktop GUI
- Save simulation output in the executable folder

Technologies Used

- Python 3.6+
- PyQt6
- OpenModelica / OMEdit
- Windows 10/11

Project Structure

```
OpenModelica_Project/  
├── main.py  
├── README.md  
├── model/  
│   ├── FlowConnect.mo  
│   ├── Tank.mo  
│   ├── Tank2.mo  
│   ├── TwoConnectedTanks.mo  
│   ├── package.mo  
│   └── package.order  
└── executable/  
    ├── NonInteractingTanks.TwoConnectedTanks/  
    │   ├── TwoConnectedTanks.exe  
    │   ├── result_0_4.mat  
    │   └── and other generated runtime files
```

How It Works

1. The OpenModelica model is opened in OMEdit.
2. `TwoConnectedTanks` is compiled.
3. OMEdit generates the executable and required runtime files.
4. The Python GUI lets the user choose the executable.
5. The user enters start and stop time values.
6. The app runs the executable with those arguments.

Input Constraint

For the screening task, the following condition is enforced:

```
0 <= start time < stop time < 5
```

Only integer values are accepted.

How to Run

1. Run the GUI

Open a terminal inside the project folder and run:

```
py -3.14 main.py
```

2. Use the GUI

- Click **Browse**
- Select the `TwoConnectedTanks.exe` file from the OpenModelica output folder
- Enter start time and stop time
- Click **Run Simulation**

OpenModelica Output Files

The executable should be used from the OpenModelica generated folder that contains the runtime files and simulation outputs. The submission should include the executable along with its dependent generated files.

Notes

- Do not use the copied executable from another folder if it is missing its dependent runtime files.
- Use the OpenModelica-generated simulation folder for reliable execution.
- The GUI validates the input range before running the simulation.

Submission Checklist

- [x] OpenModelica model compiled successfully

- ☒ Executable generated from `TwoConnectedTanks`
- ☒ Python GUI created with PyQt6
- ☒ Executable path selection added
- ☒ Start time and stop time inputs added
- ☒ Run button added
- ☒ Input validation implemented
- ☐ GitHub repository created
- ☐ Files uploaded to repository
- ☐ Submission link shared as required

Author

Created as part of the OpenModelica + Python desktop app screening task.